

Inputs

Operational records

full history of every field the auditor holds

Disclosure rules

category and release timing of each field

Field roles

candidates G , public baseline B , targets y

Set level: inferability $V_y(S)$

Amortized attacker (scan)

masked pre-training once per dataset,
closed-form readout per query
 $\rightarrow \hat{V}_y(S)$ for all $|S| \leq K+1$, ms per set

*top-k
backgrounds*

Dedicated retraining (certify)

single-/multi-target DNN, gradient
boosting; best one on validation;
test R^2 + monotone closure $\rightarrow V_y(S)$

*certified
lower bound*

Field level: risk $M_{i \rightarrow y}^{(K)}$

Budgeted marginal risk

$$M_{i \rightarrow y}^{(K)} = \max_{|T| \leq K} \Delta_i(B \cup T)$$
$$\Delta_i(A) = V_y(A \cup i) - V_y(A)$$

- single-field risk $M^{(0)}$
- amplification $\Gamma^{(K)} = M^{(K)} - M^{(0)}$
- top backgrounds T^*
- gain matrix $V(ij) - V(j)$
- τ -critical $\Leftrightarrow$ in a minimal unsafe set of size $\leq K+1$

Decisions

Grading

grade by $M^{(K)}$, flag
amplified fields

Withholding

break the minimal
unsafe sets; do not
co-release with T^*

Re-grading

recompute when the
public baseline grows